MERN Stack Notes - Part 3: MongoDB vs PostgreSQL

1. Key Differences:

- Data Model: MongoDB is document-oriented (JSON/BSON); PostgreSQL is relational (tables & rows).
- Schema: MongoDB offers flexible, dynamic schema; PostgreSQL enforces strict schemas and constraint
- Transactions: PostgreSQL provides mature ACID transactions out of the box; MongoDB added multi-doc
- Scaling: MongoDB scales horizontally via sharding; PostgreSQL typically scales vertically with read-repli

2. Indexing in MongoDB:

MongoDB uses B-Trees for indexes. Single field, compound indexes, and multikey indexes speed up queries significantly by avoiding full collection scans (COLLSCAN).